Specification Surfaces for Incremental Forecast Performance: Decomposition and Simultaneous Inference for Nested Probability Forecasts

Abstract

Forecast evaluation asks whether a predictor improves out-of-sample probability forecasts over a nested benchmark, answering with one score difference dependent on unstated choices: the benchmark, model class, split, forecast origin, predictor completeness and scoring rule. We treat the improvement as a surface over declared choices, with three reporting objects: an exact variance decomposition separating forecaster choices from split-to-split variation; simultaneous bootstrap bands, their family-wise coverage measured by simulation; and resolution regions, where its sign is determined. Across 19 binary forecasting tasks from 13 public event logs, mostly IT-service and administrative, the evaluation split and its interactions carry a median 71.7% of the variance, and resolved specifications (9 tasks) lie a median 0.0600 in area under the ROC curve from the conventional report. In a bank service-desk case study, the predictor improves reassignment forecasts at first touch, not at incident creation, once the benchmark holds what is assumed recorded by then.

Keywords: Evaluating forecasts, Error measures, Bootstrapping, Probability forecasting, Robustness, Decision making, Statistical tests, Forecasting practice

Highlights

- A predictor's incremental forecast accuracy is a surface, not one number
- An exact decomposition separates the evaluation split from other forecaster choices
- Simultaneous bands over a declared family, with coverage measured by simulation
- Resolved specifications sit a median 0.0600 AUC from the one-number report
- In service-desk forecasting, a predictor's value depends on the forecast origin

1. Introduction

Does a candidate predictor improve a forecast? Forecast evaluation answers by comparing nested models out of sample — a benchmark with and without the predictor (Clark & McCracken, 2001; Clark & West, 2007) — and reports the answer as one number, the difference in a score. That number is conditional on choices the forecaster makes and seldom reports: the benchmark information set, the model class and its encoding, the forecast target, the sample split, the *forecast origin* (the time at which the forecast is issued, which fixes what is known about a case), how completely the predictor is recorded, and the score and operating point at which the difference is read. Several of these have been studied one at a time — the estimation window (Rossi & Inoue, 2012), the sample split (Hansen & Timmermann, 2012), the accuracy measure (Kolassa, 2020) — and applied work finds that whether a predictor helps can turn on the sample and the specification that carry it (Stock & Watson, 2003; Welch & Goyal, 2008; Bańbura & Bobeica, 2023). What is missing is a way to report how a predictor's incremental forecast performance varies jointly over all of them, with uncertainty that holds jointly over a declared family of specifications. This paper supplies one for probability forecasts of binary events.

The application is service-operations forecasting. A service desk decides, as each case arrives, where to route it, and a case that is later passed to another team, or that stays open far longer than usual, costs the desk time and service levels. We forecast those two events — *reassignment* (handover to another team) before resolution, and *long duration* — as probabilities issued at a declared origin, and ask whether a field from the organisation's inventory of the things its cases concern, called here the *register*, improves the forecast over the case's arrival attributes. In IT service management that inventory is the configuration management database, and in the case study of Section 7 the candidate predictor is the configuration item an incident affects. Service desks sit beside the call centres and emergency departments whose operations forecasters already model (Ibrahim, Ye, L'Ecuyer & Shen, 2016; Rostami-Tabar & Ziel, 2022), and their case-level event forecasts are of the kind made for customer churn and credit default (De Caigny, Coussement, De Bock & Lessmann, 2020; Liu, Fan & Xia, 2023).

Our position is that the dependence on unstated choices is large enough, on real forecasting decisions, that the number alone is not a report; that the right object is a surface over a *declared* design space; and that the surface admits estimation and inference of its own. Two findings follow.

A pooled specification surface overstates the forecaster's latitude: most of its variance here comes from the evaluation split and its interactions — sampling variability and fold-to-fold instability — rather than from the forecaster's other choices, and an exact decomposition partitions the two (Section 4.4). And **where the data determine the sign of the improvement, the conventional report usually has the right sign but does not show how far other defensible specifications lie from it**; Section 6.4 measures that distance.

The estimand. Write a specification s for a complete assignment to every design axis, and

$$V_s(f) \;=\; m_\theta(A_s(D_{\text{train}}, B \cup \{f_q\}),\, D_{\text{test}}) \;-\; m_\theta(A_s(D_{\text{train}}, B),\, D_{\text{test}})\,, \tag{1}$$

where A_s is the fitting procedure (learner, encoding, hyperparameters, calibration), B the baseline feature set, f_q the field under a register quality mechanism q, m the metric and θ the operating point, and where $D_{\text{train}}, D_{\text{test}}$ are fixed by the split rule and the cohort. The *specification surface* is the map $s \mapsto V_s(f)$ over a design space $\mathcal{S}$ that the analyst declares and the reader can enumerate; Figure 1 draws one. Equation (1) measures how much better a model forecasts, not what the field costs or what acting on the forecast is worth; Section 8 converts the increment into net benefit at a declared exchange rate, the sense in which Murphy (1993) and Granger & Pesaran (2000) speak of a forecast's value.

What kind of forecast this is. Every forecast here is a case-level probability forecast of a binary event — that a case will be handed over to another team, or will stay open longer than the training period's median — issued once, at a declared forecast origin, from the information recorded by then, as in churn, credit-default and recession-probability forecasting (Liu & Moench, 2016). It is scored on a later hold-out period by 5 declared scalar metrics, the area under the receiver operating characteristic (ROC) curve first among them, and in Section 8 by net benefit at a declared operating point. The objects that follow apply to any increment in a score between two nested forecasting pipelines, with or without a time-series structure; the moving-block bootstrap that produces the bands is used because cases arrive in time order.

Contributions.

1. **An estimand for nested forecast comparisons that includes the evaluation design.** The incremental score as a function of the benchmark, the model class, the sample split,

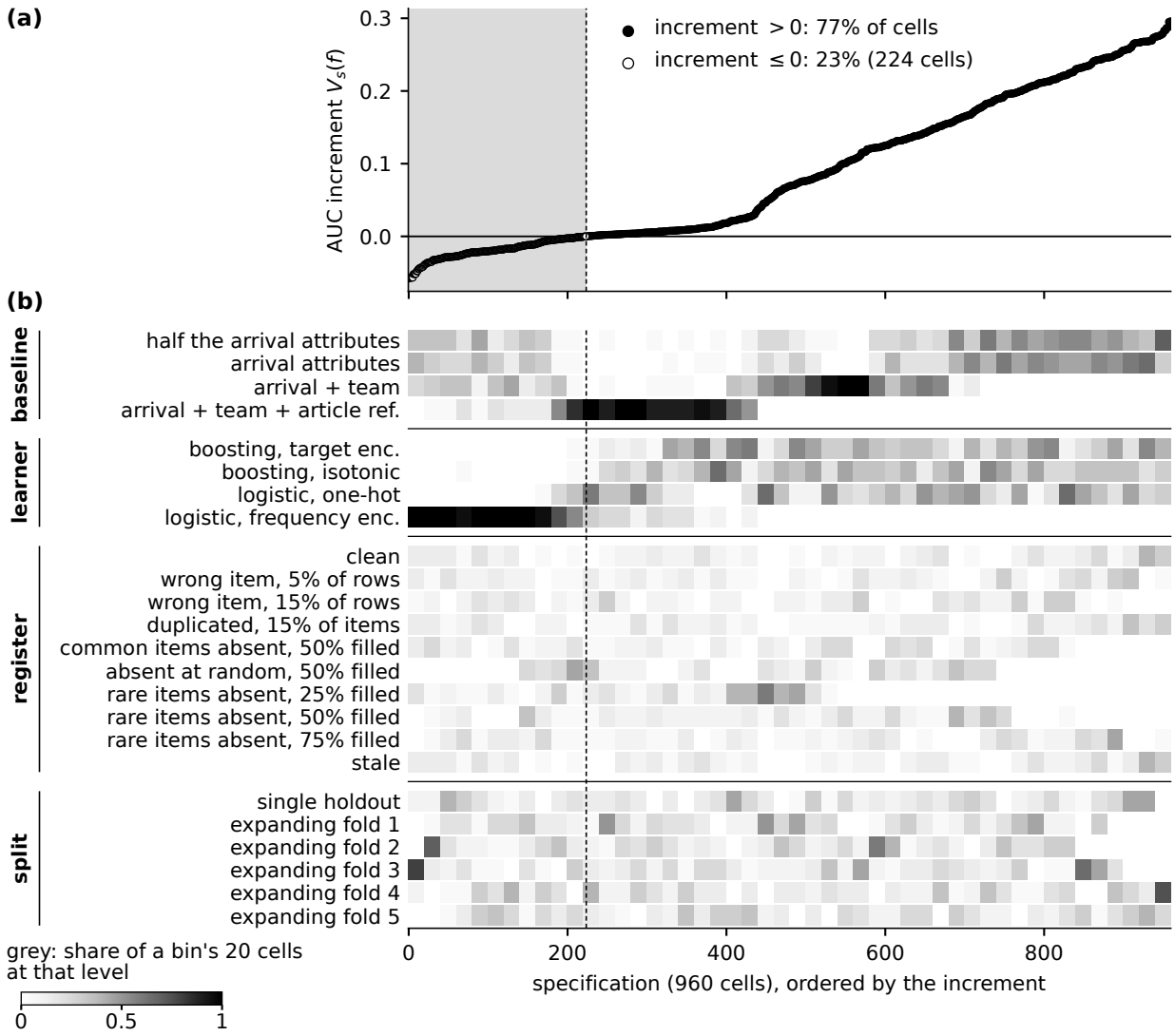


Figure 1: The specification curve for the configuration item on BPI Challenge 2014 (BPIC14), at the pre-specified handover target, in the primary metric, the area under the ROC curve (AUC): 4 pipelines × 4 nested baselines × 10 register-quality conditions × 6 splits. The other metrics are not on a common scale with AUC (Remark 1). **(a)** Each admissible specification, sorted by its increment: filled if positive, open on the shaded band if not. **(b)** For each level of each axis, the share of the cells in each sorted bin that sit at that level; the dashed line marks the sign change. The intercept-only baseline is excluded. Point estimates only; the region counts the simultaneous bands decide are in Table 3.

the forecast origin, the predictor's data quality and the scoring rule, over a design space the reader can enumerate (Table 2); and an exact functional-ANOVA decomposition of its variance in which the evaluation split is an *error stratum*, so that the evaluation split, with

its sampling variability and fold-to-fold instability, is separated from the forecaster's other choices (Section 4.4).

2. **Simultaneous inference over a declared family.** Max-t bands from a bootstrap that refits the forecasting models in every draw, at a critical value that attains 94.7% family-wise coverage in a simulation matched to this design (Section 9.2) — an upper bound, because that simulation's bootstrap is ideal — and *resolution regions*, which state where the sign of the improvement is determined, with a robustness index ρ under a declared minimum resolved share. Along the way, refitting a high-cardinality predictor inside each draw is shown in simulation to break the percentile bootstrap interval, which is why every interval here is the basic one (Section 4.1).
3. **Evidence on 19 pre-specified forecasting tasks and a service-desk case study in which the forecast origin is part of the estimand.** One frozen pipeline is run on every log–target pair a pre-specified protocol admits, not only those where the answer resolves, and every corpus statistic is reported on all 19 and on the 8 IT-service pairs (Supplement S15.15). In the case study (Section 7) the predictor's increment is large when the reassignment forecast is issued at first touch and indistinguishable from zero at incident creation once the benchmark holds the fields assumed recorded by then; a decision-based evaluation prices what reading the wrong benchmark costs a desk (Section 8.3).

Figure 2 summarises the method. *Specification regret*, what choosing one specification costs against the best, is a robustness check only; it is null on this corpus (Supplement S3.2).

Scope and supplement. This is a retrospective study on public data, offered as a forecast evaluation methodology; it is not a systematic review of reporting practice or a report of a deployed forecasting system, and Section 10 says what each would have required. The protocol and its amendments, the proofs, the correction register and the tables this article does not print are in a supplementary document whose sections, tables and figures carry an S; Supplement S10 is a glossary of every term used here in a technical sense.

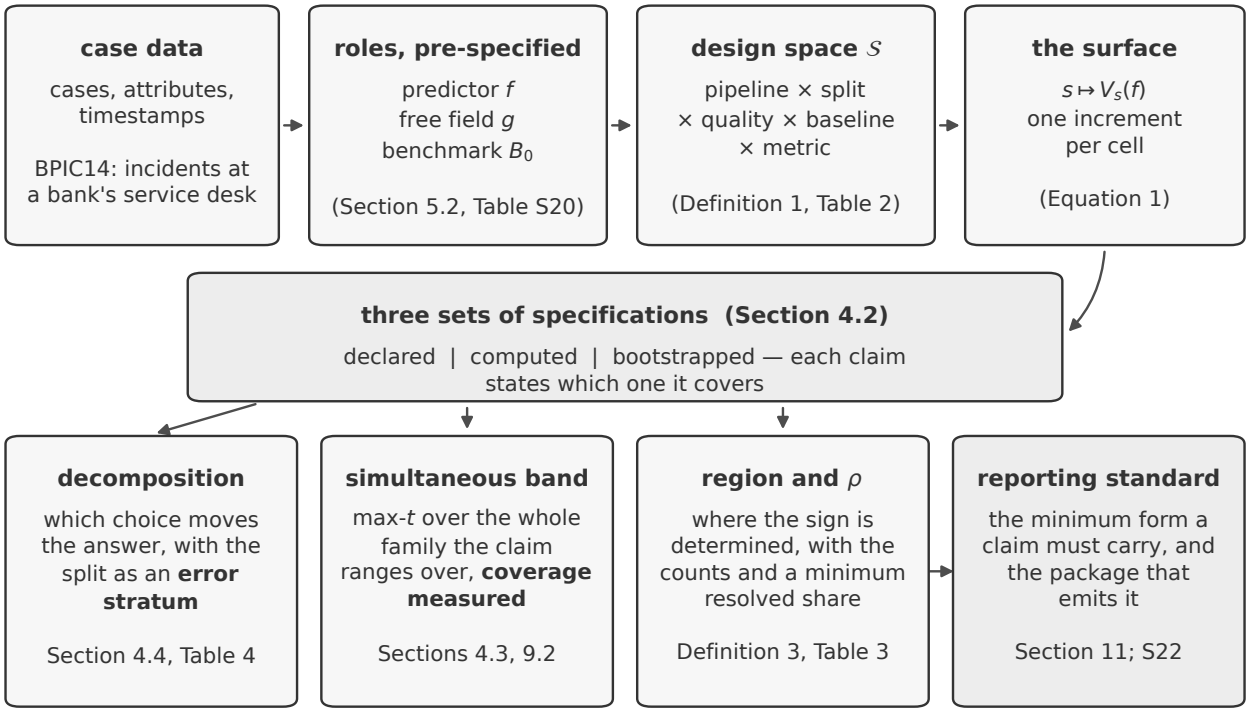


Figure 2: The method, once. A log is read into three *roles* by declared rules; the roles are crossed over a declared *design space*; the result is a *surface*, one increment per cell. Every claim about that surface names which of three *denominators* it quantifies over, and the reporting objects are computed from it. Each box names the section or table in which its object is defined and reported.

object	what it is for	§	table	what it costs
Sensitivity decomposition	how much of the disagreement each analysis choice explains, and how much belongs to no single one	4.4	S12	exact; no refit. A Möbius sum over the surface already computed
Simultaneous band	an interval that holds over every cell a claim quantifies over, not one cell at a time	4.3	S13	the whole cost: one draw refits every arm of every cell
Resolution region and ρ	which way the surface points where the data settle it, and how much of it they settle	4.5	S31	a function of the band; no further refit
Specification regret *(a check, not an object)*	what choosing one specification costs against the best one, on a scale a decision can use — it returns a null on this corpus	S3.1	S34	exact; no refit, on the surface already computed

Table 1: **The reporting objects.** The three this paper supplies and the fourth it retains as a robustness check, what each is for, where it is defined, where it is reported on this corpus, and what it costs; the two middle columns are the section it is defined in and the table that reports it. Only the second costs model fits: a draw of the bootstrap refits every arm of every cell, which is why the surface an inference is taken over is smaller than the surface a point estimate is taken over (Section 4.2). The other three are functions of a surface that has already been computed, so an analyst who has done the sweep has already paid for them.

2. Related work

In the forecasting literature the incremental question is asked of one evaluation design at a time. Whether one forecast is more accurate than another is tested by Diebold & Mariano (1995) and West (1996); whether a predictor adds accuracy to a model that nests the one without it — the incremental question, exactly — by Clark & McCracken (2001), and Clark & West (2007) show that under the null, estimating the larger model's extra parameters inflates its out-of-sample mean squared error, biasing the raw comparison against it; whether it adds accuracy *conditionally*, at a given origin and state, by Giacomini & White (2006). Gonzalo & Pitarakis (2024) extend nested out-of-sample comparison to many candidate predictors at once, and Liu & Swanson (2024) measure the marginal predictive content of predictor groups across targets and horizons; both vary the predictor and hold the evaluation design fixed, where this paper holds the predictor fixed and varies the design. Probability forecasts of a binary event are evaluated in the area under the ROC curve (Lahiri & Wang, 2013), in which a predictor's increment over a nested baseline is read (Liu & Moench, 2016) and tests of that increment are badly behaved at the null (Demler, Pencina & D'Agostino Sr., 2012; Pepe, Kerr, Longton & Wang, 2013) — one reason the increments here are reported as estimates with intervals, although deciding a sign still operates at that null — and in proper scores (Brier, 1950; Gneiting & Raftery, 2007), whose decomposition separates the calibration an AUC ignores from the discrimination it measures (Dimitriadis, Gneiting & Jordan, 2021). Reliability, ROC and Murphy curves together separate calibration, discrimination and value when probabilistic classifiers are compared (Dimitriadis, Gneiting, Jordan & Vogel, 2024); this paper evaluates the increment between two nested forecasts across a design, and Section 8 reads that increment in net benefit.

Several of the choices this paper puts on a surface have been studied one at a time: the estimation window, over which Rossi & Inoue (2012) make tests robust; the sample split, whose choice Hansen & Timmermann (2012) show can manufacture rejections, and whose place in the limit theory of nested comparisons Hansen & Timmermann (2015) characterise through Wald statistics; the forecast origin, along which Giacomini & Rossi (2010) track relative performance and which rolling-origin evaluation varies by design (Tashman, 2000); and the accuracy measure, which can change which forecast is best (Hyndman & Koehler, 2006; Kolassa, 2020). Applied forecasters meet the joint dependence: whether the Phillips curve helps forecast euro-area inflation

turns on the specification chosen, and averaging over modelling choices helps (Bańbura & Bobeica, 2023). Petropoulos et al. (2022) and Hewamalage, Ackermann & Bergmeir (2023) collect the evaluation practice these rest on. When the comparison ranges over a *family* of specifications, the reality check (White, 2000), the test for superior predictive ability (Hansen, 2005) and the model confidence set (Hansen, Lunde & Nason, 2011) control the search: the first two ask whether the best member of a family beats a benchmark once the search is accounted for, the third returns the set of members not rejected as inferior at a declared level; and Hunt (2022) shows that pseudo-out-of-sample tests do not keep a stable family-wise error rate when models are searched for. This paper sits beside the family tests and takes a different object from them: not whether the family contains a member that beats the benchmark, nor which members survive, but the shape of a fixed predictor's increment across the family — how much of its variation is the forecaster's and how much is resampling, where its sign is determined, and how far its members sit from the one that would conventionally be reported. A forecaster's usual answer to many specifications is to combine them (Wang, Hyndman, Li & Kang, 2023); this paper asks the prior question of how much they disagree, and where. Forecasting also has precedents for working over the specification space rather than at a single specification: recursive selection over every subset of candidate regressors at each forecast origin, to ask whether predictability survives the search (Pesaran & Timmermann, 1995), and thick modelling, which keeps and combines the forecasts of many specifications instead of selecting one (Granger & Jeon, 2004; Aiolfi & Favero, 2005). Both use the space to produce a forecast; this paper uses it to report how an increment varies across it.

What a forecast is worth to a decision, as distinct from how accurate it is, is the subject of Murphy (1993) and Granger & Pesaran (2000), and the with-versus-without question is asked of many objects: of a dataset, through a betting market (Maymin, 2019); of a new data source in customer-churn forecasting, in accuracy and in campaign profit (De Caigny et al., 2020); of a member of a forecast ensemble, by leave-one-out and Shapley differences (Kim, Ray & Reich, 2026); of individual predictors, whose contributions to out-of-sample loss a performance-based Shapley value attributes (Borup, Goulet Coulombe, Rapach, Schütte & Schwenk-Nebbe, 2022); and of a step in a forecasting process, as forecast value added (Fildes, Goodwin & De Baets, 2025). Each answers under one evaluation design; none asks how the answer moves when the design moves. The clinical-prediction literature has argued for two decades that the increment in the area under

the ROC curve from adding a marker depends on the existing model and is a poor guide to usefulness (Pencina, D'Agostino Sr., D'Agostino Jr. & Vasan, 2008; Cook, 2007; Kerr, McClelland, Brown & Lumley, 2011), and decision curves read net benefit across an exchange rate (Vickers & Elkin, 2006; Vickers, Van Calster & Steyerberg, 2016), which is why the operating point is an axis here. Multiverse and specification-curve analysis (Steegen, Tuerlinckx, Gelman & Vanpaemel, 2016; Simonsohn, Simmons & Nelson, 2020) supply the sweep over analytic choices from which the robustness index below is derived. Incompletely recorded inputs are a forecasting problem in their own right (Wen, Pinson, Gu & Jin, 2024); the register-quality axis puts the predictor's completeness inside the estimand rather than in preprocessing. And the forecast origin turns what machine learning treats as leakage (Kaufman, Rosset, Perlich & Stitelman, 2012; Kapoor & Narayanan, 2023) into a declared choice of information set. Variable importance and data valuation, underspecification and predictive multiplicity, many-analysts studies, and the business-process analytics literature from which the data come, where such case-level forecasts are called outcome-oriented predictive monitoring (Teinemaa, Dumas, La Rosa & Maggi, 2019), are discussed in Supplement S1.1.

What is new for forecast evaluation. Forecast-comparison tests either fix the evaluation design and ask whether a predictor, or the best of a family of models, improves accuracy (Clark & McCracken, 2001; Hansen, 2005; Gonzalo & Pitarakis, 2024), or robustify one design choice at a time (Rossi & Inoue, 2012; Hansen & Timmermann, 2012). This paper reports how a fixed predictor's incremental accuracy varies jointly across the design, separates the evaluation split's share of that variation from the forecaster's other choices, and gives inference that holds jointly over a declared family. Three things are new. The estimand is declared completely: the forecast origin and the predictor's data quality are arguments of it rather than preliminaries, and the measure over the design is stated rather than implied by whichever cells were enumerated, which moves the answer (Section 6.2). The decomposition treats the evaluation split as an error stratum and so partitions the forecaster's choices from the split's exactly; separating specification-induced from sampling variation has precedents — extreme-bounds analysis, which reports an estimate's range across specifications (Leamer, 1983), multimodel analysis that attributes an estimate's variation to each modelling choice (Young & Holsteen, 2017), functional ANOVA over a configuration space (Hutter, Hoos & Leyton-Brown, 2014), and the error strata of classical experimental design —

and what is added is its exact application to a nested out-of-sample forecast comparison, with the evaluation split as the stratum. And the share of specifications with an improvement in a declared direction, which is standard specification-curve practice (Simonsohn et al., 2020) and which the robustness index ρ counts, is taken under a *simultaneous* band over a declared family rather than a pointwise test, over a counted set of specifications, and subject to a minimum resolved share below which no direction is reported; Section 6.5 and Supplement S18 measure where a specification curve's verdict and a region label part company, and Section 6.4 measures how far the conventional report sits from the resolved specifications. The contribution is not the observation that a single number is usually reported. It is that an incremental-performance claim is a functional of a declared design, that the functional can be estimated with quantified uncertainty, and that on this corpus it varies widely.

3. The specification surface

This section defines the object. A forecaster comparing two nested forecasting pipelines — one with the candidate predictor, one without — reports the difference in a score; the surface is that difference as a function of every choice the comparison rests on, and the design space is the set of those choices, declared before the comparison is run.

3.1. The design space

Definition 1 (Design space)**.** *A* design space $\mathcal{S}$ *for an incremental-performance claim is a finite set of specifications, each a complete assignment to the axes*

$$s = (\underbrace{A}_{\text{learner}}, \underbrace{E}_{\text{encoding}}, \underbrace{Y}_{\text{target}}, \underbrace{\Pi}_{\text{split}}, \underbrace{\tau}_{\text{forecast origin}},$$
$$\underbrace{q}_{\text{register quality}}, \underbrace{B}_{\text{baseline}}, \underbrace{m}_{\text{metric}}, \underbrace{\theta}_{\text{operating point}}),$$

together with the cohort and eligibility rule that fix D. A claim about a feature is well posed *only relative to a declared $\mathcal{S}$.*

The baselines are nested, $B_0 \subset B_1 \subset \cdots$, and we call them the *nested baselines*: B_0 is the intake block (the low-cardinality attributes recorded when a case opens), and a richer baseline adds the *free field* g, a routinely recorded attribute — on the case-study log the assignment group — that

axis	levels	in the product	measure	exclusions
learner	2–4	yes	equal / ref. / conc.	none
encoding	2–3	no: in the learner	set by the learner	none
split	6	yes	equal / ref. / conc.	none
forecast origin (availability)	1–2	no: second surface	never averaged	two origins on the primary log, one elsewhere
register quality by severity	6–10	yes	equal / ref. / conc.	none
baseline (nested, all)	4–5	no: the row below	equal / ref. / conc.	intercept-only: in the surface, out of every admissible set
baseline, admissible only	3–4	yes	equal / ref. / conc.	none
scalar metric	5	yes	never averaged across	none
operating point (net benefit)	31	no: decision curve	declared threshold dist.	outside the declared range

Table 2: The declared design space. One row per axis: the number of levels (a range where it differs by log), whether the axis is a factor of the admissible cell count, the measure placed on its levels, and every level excluded by declaration. Only the rows marked 'yes' multiply: learner × split × register quality × admissible baseline × scalar metric equals the 'declared scalar' column of Table S18 on every pair. Each row marked 'no' is counted where the column says; Supplement S1.2 explains those rows and the three measures.

may carry the same information as the register f. The metric axis has 5 levels, the *metrics*: the area under the ROC curve (AUC), average precision, the Brier skill score, the log-loss skill score and Nagelkerke's R^2, of which the Brier and log-loss skill scores are affine in a strictly proper score on a fixed test set. We call the value $V_s(f)$ at one specification a *cell* of the surface; a *scalar* cell is one whose metric is one of these scores rather than a point of the decision curve, and a cell is *admissible* if it lies in the declared space and no declared rule excludes it (for scalar cells the one such rule excludes the intercept-only baseline).

Declaring a design space means writing down every axis, its levels, the measure over those levels, and every level deliberately excluded with the reason. Table 2 is that declaration. A pair's admissible cell count is the product of five of its rows — learner, split, register-quality condition, admissible baseline and scalar metric; the note to the table says how the other axes enter. Not all nine axes are varied on every pair (Section 5 and Section 10). The measure over the levels is declared rather than left implicit, and the pooled decomposition is reported under three product measures (Table S33; Supplement S1.2).

3.2. What the surface estimates

Equation (1) is written on a fixed D_{train} and D_{test}, so as it stands $V_s(f)$ is a deterministic functional of the log in hand and an interval for it would cover with probability zero or one. The quantity the intervals of Section 4 target is the following.

Definition 2 (The estimand)**.** *Let the cases of a log be a sample from a population P, and let the specification s fix the fitting procedure A_s, the cohort and eligibility rule, the split rule Π, the forecast origin, the register-quality mechanism, the baseline and the metric. Write $A_s(P)$ for the limit of the fitted score function as the sample grows under P with s held fixed. The limit requires the case sequence to be stationary and mixing, so that a temporal split does not separate two populations and the moving-block bootstrap of Section 4.1 resamples a persistent dependence structure; stationarity is tested and does not hold everywhere — the increment differs across the test half on 6 of 19 pairs (Supplement S13) — and mixing is not tested (Supplement S21). The* estimand at s *is*

$$\vartheta_s(f) \;=\; m_\theta\big(A_s(P, B \cup \{f_q\}),\, P\big) \;-\; m_\theta\big(A_s(P, B),\, P\big),$$

and $V_s(f)$ of (1) *is its plug-in estimate on the log.*

Three things follow. The split rule and the pipeline are *part of the estimand*, not nuisances to be averaged away, which is why the nested bootstrap of Section 4.1 resamples the training half rather than conditioning on it. Under misspecification ϑ_s is not what the register is worth in the world; the simulation of Section 9 separates the two as V_{limit} and V_{oracle}. And every cell of every table is an *estimate* of ϑ_s, so "resolved" in Definition 3 means that the band excludes zero for ϑ_s, not that the observed increment is nonzero.

3.3. The surface, and why it is not a scalar

Given $\mathcal{S}$ and data D, the surface is the map $s \mapsto V_s(f)$ of (1). The *conventional report* is one cell of it, the reference cell: one-hot logistic regression, the single temporal holdout, the clean register, the intake block plus the free field, and AUC. The paper's claim is that a single cell is not a summary of the surface in any useful sense. The point estimates alone establish that: on 18 of the 19 pairs, between a tenth and nine tenths of the admissible cells are positive. Whether that amounts to a *misstatement* is a harder question, which Section 6.4 answers with three restrictions rather than with a rate. A scalar is defensible in one state: when the whole admissible surface lies on the same side of zero with simultaneous confidence, a positive number is a lower bound on a robust conclusion (Section 4.5).

Comparisons between two baselines are often stated as a proportion. Write $B_0 \subset B_1$ and

$$D_s(f) \;=\; V_s(f \mid B_0) - V_s(f \mid B_1), \quad R_s(f) \;=\; 1 - \frac{V_s(f \mid B_1)}{V_s(f \mid B_0)} \;=\; \frac{D_s(f)}{V_s(f \mid B_0)}; \tag{2}$$

$D_s(f)$, the *absorption*, is how much of the register's increment the richer baseline B_1 takes up, in the metric's own units. R is a ratio and inherits every pathology of one, so D and the two increments are primary and R is a secondary descriptive statistic, reported under three declared rules with a Fieller set (Fieller, 1954) whose *kind* — bounded, unbounded, or the complement of an interval — is printed beside it (Supplement S20.1 and Supplement S3).

Remark 1 (The metric is an argument of the estimand)**.** *No function carries the AUC reduction to the average-precision reduction: two configurations can share an AUC reduction to* 2.2×10^{-16} *while their average-precision reductions differ by 0.417. The reduction is therefore not identified by the data, the feature and the two baselines, which is why every table in this paper names its metric and why the minimal practically important difference of Section 6.4 is declared in one metric's units and applied there only. That forecast rankings can depend on the scoring function is established for point and quantile forecasts (Ehm, Gneiting, Jordan & Krüger, 2016; Patton, 2020); the remark is its counterpart for an increment between nested models.*

A second property — Remark S1, that a reduction is invariant to a declared family of recalibrations exactly when the metric is *affine* under it — is stated and proved in Supplement S2.

4. Estimation and inference

This section gives the estimator of one cell and its interval, then the objects that read the surface as a whole: the set a claim ranges over, simultaneous bands, the variance decomposition, and the regions where a sign is determined.

4.1. The estimator, and uncertainty that includes the model

The pipeline is frozen and identical on every log, and the *pipeline* axis spans encodings as well as model families; all models are fitted with `scikit-learn` (Pedregosa, Varoquaux, Gramfort, Michel, Thirion, Grisel et al., 2011). Attributes are assigned to roles by the pre-specified rules of Section 5, cases are ordered by their first timestamp, and the split is temporal (Supplement S3.3). The usual interval for such an increment resamples test rows with the models held fixed, which excludes training-sample, model-selection, encoder and split variability and the temporal dependence between cases. Every interval here is instead a *nested moving-block bootstrap*: one draw resamples the training half by moving blocks (Künsch, 1989) of length $\lceil n^{1/3} \rceil$, refits both *arms* of

every comparison — the model with the register and the nested model without it — under every baseline, learner and quality mechanism, and evaluates on a moving-block resample of the test half, with encoders, frequency tables, target encodings, register rankings and calibrators rebuilt inside the draw. The nested construction is used because the variability it captures is part of the estimand, not because it covers better: in the simulation of Section 9 the two intervals cannot be told apart at its Monte Carlo precision.

Throughout, K denotes the **cardinality of the register field** — the number of distinct values f takes on a pair, 337 at the median pair and 5,245 at the largest (Table 3) — and n the number of training cases. The ratio K/n governs everything in this subsection and in Section 6.3.

Refitting inside the draw breaks the percentile interval. Refitting every arm inside every draw displaces the bootstrap distribution from the estimate. Part of the cause is that a multinomial resample holds only about $1 - e^{-1}$ of a high-cardinality categorical's distinct levels, so a refit sees a sparser register, but removing that loss leaves most of the displacement standing (next paragraph). A percentile interval taken from a displaced distribution is a correct interval for the wrong quantity. On the sparse world of Section 9 it covers 37.2% against a nominal 95%, *worse* than the fixed-model interval refitting was meant to improve on. Every interval here is therefore the *basic* (pivotal) interval $[\, 2\hat{V} - q_{1-\alpha/2},\ 2\hat{V} - q_{\alpha/2} \,]$ (Davison & Hinkley, 1997), and simultaneous bands are recentred by the same shift (Supplement S3.8).

Weights rather than subsets. Every band comes from a bootstrap that draws Dirichlet *weights* per moving block rather than a subset of rows, so every register level is present in every refit and no cell loses its band because a draw dropped an arm — 0 unbandable cells under weights, against 270 of 3,420 under a multinomial resample of the same design. Weights do not remove the displacement of the bootstrap distribution; its residual cause is not identified here and is listed among the limitations (Supplement S3.9 and Supplement S21). One candidate mechanism, not established here, is that block weights lower the effective training size, so a draw sits lower on the learning curve that K/n governs than the full-sample estimate does.

The analysis code, and separate code that re-derives every reported number, were written with the help of a generative-AI tool (see the declaration before the references); no reported number depends on the tool being run again, because the archived code regenerates every result from the public logs (see the data availability statement).

4.2. Three surfaces, and which one a claim may quantify over

A statement of the form *this surface is uniformly beneficial* ranges over a set of specifications, and three sets are distinguished and counted (Table S18). The **declared surface** is every admissible specification: the product of the axis levels declared in Section 5. The **computational surface** is every specification actually evaluated; here it equals the declared surface: all 29,040 declared scalar cells were evaluated, with 0 missing and 0 duplicated. The **inference surface** is every specification carrying bootstrap draws; it is smaller because a draw refits the whole pipeline, so its cost is the declared surface's multiplied by the draw count.

The rule that selects the inference surface reads neither a result nor the log: one design on every pair — two learners, two splits, three register-quality conditions and three admissible baselines, crossed, 36 cells a pair per metric and 3,420 scalar family members across the corpus, at 400 draws — a complete factorial, identical on every pair (Supplement S3.3; Supplement S15.2 compares the decomposition on the two surfaces). The inference family is a median 16.7% of a pair's admissible cells, and less than a twentieth on the pair with the richest grid; **a label computed over fewer cells is systematically cleaner than one computed over all of them** (Section 4.5). Only the region labels use the inference surface; the decomposition, the sign-disagreement rate (the share of cells whose sign differs from the conventional report's) and the regret comparison use the full declared surface.

4.3. Simultaneous bands over the family the claim ranges over

A decision curve read across 31 thresholds, a set of nested baselines read together, or a surface read across every admissible cell, is a family of statements, and a pointwise interval does not control the probability that *some* member of the family is wrong. For a declared family $\mathcal{F}$ of cells with bootstrap draws $V_c^{(b)}$ and per-cell standard errors $\hat{\sigma}_c$, let

$$T^{(b)} \;=\; \max_{c \in \mathcal{F}} \frac{\left|V_c^{(b)} - \bar{V}_c\right|}{\hat{\sigma}_c}, \qquad q_{1-\alpha} = \text{the } (1-\alpha) \text{ quantile of } \{T^{(b)}\},$$

and report $\hat{V}_c \pm q_{1-\alpha}\hat{\sigma}_c$, recentred by the basic construction's shift as above. This is the max-t (Romano–Wolf) construction (Romano & Wolf, 2005) and has family-wise coverage over $\mathcal{F}$ asymptotically, and over nothing larger. It differs from the model confidence set (Hansen et al., 2011) in what it returns: the confidence set is the subset of a family not rejected as inferior to its best member, a statement about *which* specifications survive; the band is a simultaneous interval for

every member's increment, from which a region label reads where the sign is determined and a robustness index how much of the family agrees. Both rest on a bootstrap of the maximum over the family; the objects read off it are different.

Which family a band must cover. Within-cell bands, each covering the 5 metrics of one cell, do not cover the several hundred cells that a surface-level label ranges over. The WHOLE-SURFACE family — max-t over *every* scalar cell of the pair's inference family — is the only family a surface-level claim may use; it has 180 members on every pair against the within-cell family's 5, at a median critical value of 5.03 against that family's 2.37 and the pointwise 1.96. Family control is per log–target pair, so every cross-pair statement here is descriptive; Supplement S3.4 defines the four interval objects and the rule for dropping a replicate that leaves a cell without outcome variation.

Estimating the critical value. Two estimators of $q_{1-\alpha}$ are computed on every family. The first is the empirical $(1-\alpha)$ quantile of the B observed maxima — the Romano–Wolf construction as cited — whose precision is that of an order statistic of B values. The second multiplies the standardised draws by independent standard normals (Chernozhukov, Chetverikov & Kato, 2013), which frees the precision of q from B but reproduces only the excursions a Gaussian process makes; on every family here it lies below the empirical quantile's whole order-statistic interval (Supplement S3.6). **The operative critical value is the empirical quantile**; the multiplier's median on the whole-surface families is 3.36, and Section 9.2 measures the family-wise coverage of both against a known answer. A max-t band also inherits the coverage of the pointwise interval it is built from, which Section 9.1 shows is governed by K/n; Supplement S9 derives a widening at each pair's own ratio and reports it as a sensitivity.

Five contrasts on the primary log were fixed before the final analysis but after earlier analyses of the same data, so their Holm-adjusted p-values are descriptive (Supplement S3.5); every other comparison in this paper is exploratory and carries no multiplicity correction.

4.4. The sensitivity decomposition

Because the declared surface is a complete factorial (Section 4.2), the variance of V_s admits an exact, orthogonal functional-ANOVA decomposition (Sobol', 2001; Saltelli, Annoni, Azzini, Campolongo, Ratto & Tarantola, 2010) under a product measure $\mathbb{P} = \bigotimes_i w_i$, with first-order and total indices S_i and S_{T_i} computed exactly rather than estimated (Supplement S3); unlike the

configuration-space decomposition of Hutter et al. (2014), the axes here are analysis decisions and the components are exact rather than surrogate estimates.

The split is an error stratum, and the primary decomposition treats it as one. Five of the split axis's six levels are expanding-window folds on the same data. They are different *samples*, not different analyses, and a decomposition that pools them reports sampling variability as specification sensitivity. We call the stratum *resampling* for brevity; because the folds' training sets grow and their test periods move, it also carries the time structure of the evaluation window — the instability Giacomini & Rossi (2010) and Rossi & Inoue (2012) address — and Section 6.2 separates the split's main effect from its interactions. Because the decomposition is exact and orthogonal, the repair is a partition rather than a model: every component either involves the split or does not, so

$$1 = \underbrace{\textstyle\sum_{\emptyset \neq u \subseteq \mathcal{A}} S_u}_{\text{analyst choice}} + \underbrace{S_{\{\text{quality}\}}}_{\text{counterfactual}} + \underbrace{\textstyle\sum_{u \ni \text{quality},\, u \cap \mathcal{A} \neq \emptyset,\, u \not\ni \text{split}} S_u}_{\text{mixed}} + \underbrace{\textstyle\sum_{u \ni \text{split}} S_u}_{\text{resampling}}, \tag{3}$$

where $\mathcal{A} = \{\text{pipeline}, \text{baseline}\}$, the axes are $\mathcal{A}$, the register-quality condition (which alters the state of the world rather than the analysis) and the split, and the third term collects the split-free components that join the register-quality condition to an analyst axis; the four terms are computed exactly from the point estimates (Table 4) — exact as arithmetic, though the estimates they partition carry sampling error — and the identity holds to 4.4×10^{-16}. The first three survive fold-averaging and are decomposed again on the fold-averaged surface, where the higher-order share is net of resampling. **Section 6.2 reports that decomposition as the primary one**, and the pooled one beside it.

The share carried by interactions is

$$S_{\text{int}} \;=\; 1 - \sum_i S_i, \tag{4}$$

which is not the largest per-axis involvement $\max_i(S_{T_i} - S_i)$, because a two-way component appears in both its axes' total effects (Supplement S15.10).

4.5. Resolution regions and the robustness index

Definition 3 (Cell labels and surface regions)**.** *Let $\mathcal{F}$ be the declared inference family of Section 4.2 for a pair — the cells that carry bootstrap draws. Under the simultaneous band of Section 4.3, a cell of $\mathcal{F}$ is* beneficial *if its lower band is above zero,* harmful *if its upper band is below zero, and*

unresolved *otherwise. A surface is* uniformly beneficial *if every cell of* $\mathcal{F}$ *is beneficial;* conditionally beneficial *if some are and none is harmful;* uniformly harmful *and* conditionally harmful *in mirror image;* sign-changing *if there is at least one cell of each kind; and* unresolved *if none is resolved — **or if fewer than 5% of* $\mathcal{F}$ *is resolved at all**. The* robustness index *is*

$$\rho \;=\; \frac{\#\{\textit{beneficial}\} - \#\{\textit{harmful}\}}{|\mathcal{F}|} \;\in\; [-1, 1],$$

and every directional label is reported with the counts of beneficial, harmful and unresolved cells it rests on.

Here a cell is *resolved*, and a region a *resolution* region, when the band determines a sign; the word is unrelated to the resolution component of the Brier score (Murphy, 1973).

The minimum resolved share is a declared convention. Without it a surface earns a direction from as few as two resolved cells of a hundred and eighty, and ρ alone cannot distinguish two resolved cells from ninety, so the counts are printed beside it. The declared threshold of 5% withdraws the direction on 3 of the 9 pairs that carry one, against 0 at 1% and 5 at 10%. Because the inference family is a subset of the admissible surface, a label computed on it is cleaner than one computed on every cell, and *uniformly beneficial* gains the most; Section 6.3 therefore reports, for the one pair that attains it, the share of its admissible cells the label covers.

Before the minimum-share rule is applied, 2 surfaces are conditionally harmful and 0 uniformly harmful; each conditionally harmful label rests on at most 4 resolved cells of 180, so the rule withdraws each, and 0 harmful labels of either kind remain (Table S31). $\rho = 1$ is the only state in which one positive number is a safe summary of a surface. The region is defined over the *scalar* family; the decision curve gets its own region (Section 8), so that a single extreme threshold cannot decide the label of a whole surface. The intercept-only baseline is fitted but excluded from every admissible set, because an increment measured against a model with no features is inflated by construction.

Specification regret — what choosing one specification costs against the best one — is computed from the same surface but reported only as a robustness check, because on this corpus it returns a null (Supplement S3.2; its reading in net benefit is in Section 8.3).

5. The pre-specified benchmark design

Which datasets enter, which recorded attribute plays the predictor, which the forecast target and which the baseline were fixed by written rules before any result on a new log existed.

5.1. What was fixed, and when

The admission rules and targets were fixed in a dated protocol before any result on a new log existed, and six later amendments are logged with their causes and the results they changed (Supplement S1). **This is not a timestamped third-party registration**: the protocol is a file in a repository whose history is under the analyst's own control, so a claim that a rule was fixed in advance is no stronger than that repository. What the design buys is that the rules are legible, dated and amendable only on the record.

5.2. Roles

A log is admitted if the pre-specified rules can find three things in it without reading an outcome.

The register f. A high-cardinality attribute that is not resource-like, has at least 50 distinct values, is reused across at least two cases on average, and — by amendment 1 — whose values recur across the temporal split, so that a document number is not mistaken for a maintained entity.

The free field g. A resource-like attribute (on the case-study log, the assignment group), present on at least 50% of cases, whose value varies within a case. Only the *first* value it takes on a case is used as a feature, while the handover target is defined from the whole sequence of values — so the feature is what is known when the case opens, the target is what happens afterwards, and the field cannot inform the model about its own future.

The intake block B_0. Every remaining attribute of cardinality at most 200 that is neither an outcome, nor a timestamp, nor a relabelling of the activity, nor a deterministic coarsening of f.

The nested baselines are built from these roles: the intercept-only model; half of the intake block (its lower-cardinality half, fixed by a deterministic order before any outcome is seen); the whole intake block; the block plus the free field g; and, where a log admits a further field, that field

as a baseline of its own — on the primary log the knowledge reference, a pointer to a knowledge-base article. The intercept-only baseline is in the surface and excluded from every admissible set (Table 2).

Table S20 gives, per pair, the attribute playing each role and the number of levels each axis carries; Table 3 repeats each pair's admitted case count and register cardinality K. The *target* indexes surfaces rather than varying within one, and the per-pair factorial is 5 axes wide, not the nine of Definition 1: the forecast origin varies only on the case study's log, the operating point only in the decision-curve family, and the cohort only in Section 7.1 (Section 10).

5.3. Targets, splits and exclusions

Two targets are defined for every log — *handover*, whether more than one resource group touches the case, and *duration*, whether it runs longer than the training half's median. Cases are ordered by first timestamp and split 70/30, with five expanding-window folds beside it, and fields determined only at closure are excluded by name (Kaufman et al., 2012; Kapoor & Narayanan, 2023; Weytjens & De Weerdt, 2022). A pair is excluded only for a pre-specified reason, and every admitted pair is analysed and reported — running the surface only where the reduction resolves would condition the analysis on the outcome. Supplement S1.3 gives the rules and the exclusion codes, and Supplement S1.4 the admission margin. The case-study log's handover pair sits at the upper edge of the prevalence window, at 0.927 against a window that closes at 0.95, and carries 4,800 admissible cells, so every headline is also reported on the IT service management (ITSM) family (Supplement S15.15). The corpus is 13 public logs spanning 6 domains: a *benchmark of specification sensitivity across heterogeneous forecasting problems*, not a replication of one finding, so no effect magnitude is averaged across domains (Supplement S17).

6. Multi-log results

This section reports the surface on every admitted pair: what each analyst choice is worth, where a sign is determined, and what the conventional one-number report costs. All point estimates come from one pass of the frozen pipeline over the 19 log–target pairs; intervals are the basic construction of Section 4.1 and bands the whole-surface construction of Section 4.3.

log	target	cases	K	V at ref.	simultaneous	b/h/u	region	resolved share	ρ	sign disagr.
BPIC13_closed	handover	1,487	337	−0.009	[−0.074, +0.052]	0/0/180	unresolved	0.0	0.000	0.284
BPIC13_incidents	duration	7,554	704	0.028	[−0.016, +0.081]	12/0/168	conditionally beneficial	6.7	0.067	0.393
BPIC13_incidents	handover	7,554	704	0.037	[+0.015, +0.062]	72/0/108	conditionally beneficial	40.0	0.400	0.111
BPIC14	duration	46,606	3,019	0.096	[+0.079, +0.119]	180/0/0	uniformly beneficial	100.0	1.000	0.106
BPIC14	handover	46,606	3,019	0.167	[+0.093, +0.250]	171/0/9	conditionally beneficial	95.0	0.950	0.244
BPIC15_1	duration	1,199	89	0.001	[−0.103, +0.098]	0/0/180	unresolved	0.0	0.000	0.423
BPIC15_1	handover	1,199	89	0.007	[−0.033, +0.049]	0/0/180	unresolved	0.0	0.000	0.242
BPIC15_3	duration	1,409	94	0.035	[−0.055, +0.126]	2/0/178	unresolved*	1.1	0.011	0.191
BPIC15_3	handover	1,409	94	−0.003	[−0.126, +0.122]	0/4/176	unresolved*	2.2	−0.022	0.616
BPIC15_4	duration	1,053	53	0.035	[−0.055, +0.147]	0/0/180	unresolved	0.0	0.000	0.623
BPIC15_4	handover	1,053	53	0.001	[−0.093, +0.106]	0/0/180	unresolved	0.0	0.000	0.618
BPIC15_5	duration	1,156	107	0.032	[−0.146, +0.211]	0/0/180	unresolved	0.0	0.000	0.328
BPIC17	duration	31,509	701	0.002	[−0.018, +0.026]	0/0/180	unresolved	0.0	0.000	0.353
BPIC19	duration	251,734	1,975	0.024	[−0.020, +0.071]	10/71/99	sign-changing	45.0	−0.339	0.360
Helpdesk	duration	4,580	394	−0.020	[−0.069, +0.021]	0/2/178	unresolved*	1.1	−0.011	0.257
RoadFines	duration	150,370	66	0.031	[+0.013, +0.050]	52/6/122	sign-changing	32.2	0.256	0.173
Sepsis	duration	1,050	147	0.153	[+0.059, +0.248]	102/0/78	conditionally beneficial	56.7	0.567	0.056
UCI498	duration	24,918	5,245	−0.007	[−0.023, +0.015]	14/0/166	conditionally beneficial	7.8	0.078	0.493
UCI498	handover	24,918	5,245	−0.003	[−0.012, +0.004]	0/0/180	unresolved	0.0	0.000	0.481

Table 3: The master table. One row per admitted log–target pair: the admitted case count and the register's cardinality K; the increment V at the reference cell with that cell's whole-surface simultaneous band at the empirical critical value (Section 4.3); the counts of beneficial, harmful and unresolved cells in the inference family (b/h/u); the region label of Definition 3, minimum resolved share included ('resolved share' is the per cent of the family the band resolves); the robustness index ρ; and the share of admissible specifications whose sign disagrees with the conventional report, under the equal-level measure; 'unresolved*' is unresolved below the minimum resolved share. No (n, K) widening is applied (Table S13).

6.1. The master table

Table 3 has one row per log–target pair: the increment at the reference cell — the conventional report of Section 3 — with its simultaneous band, the counts of beneficial, harmful and unresolved cells, the region label, the robustness index and the sign-disagreement rate. The reference cell's sign and the region label need not agree, and on several pairs they do not: a surface can be conditionally beneficial while the one cell an analyst would have stood on is negative, because the label ranges over the whole inference family and the reference cell is one member of it.

6.2. What the choices are worth

Each entry of Table S11 is a median over the 5 metrics, computed within each metric in its own units. The pooled surface's largest first-order index at the median pair, 28.7%, is a different

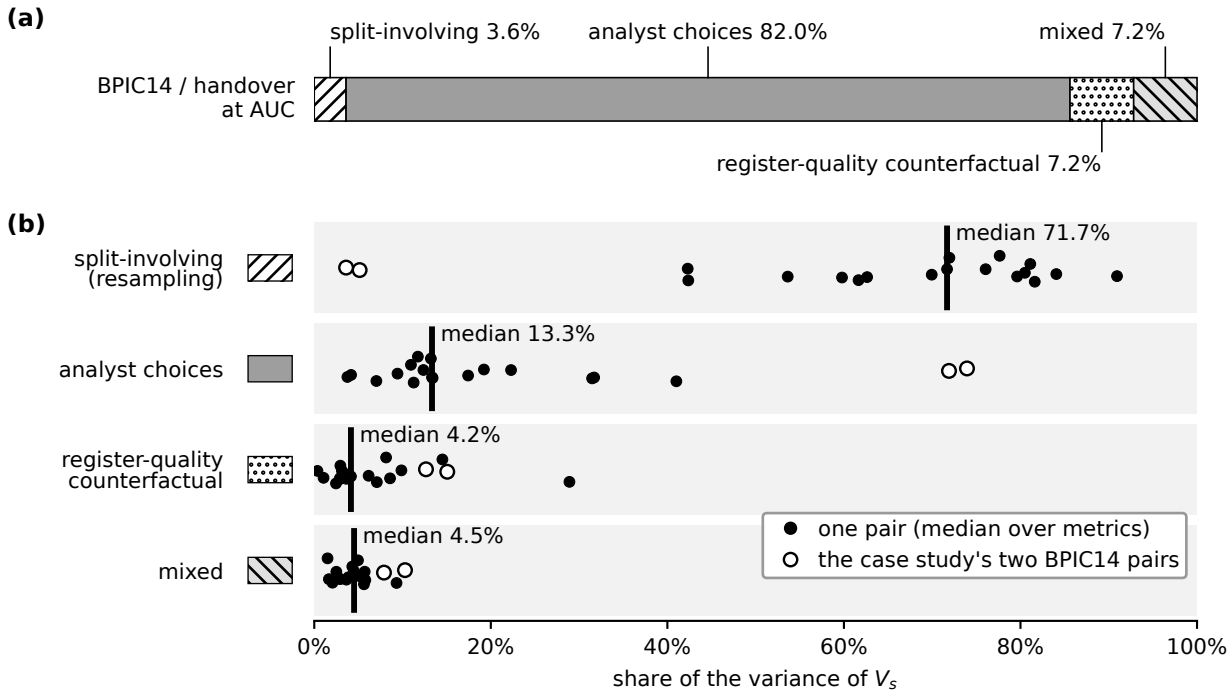


Figure 3: The decomposition with the split as an error stratum (3), under the equal-level measure. **(a)** One decomposition, so the four strata partition its variance exactly: the case-study pair BPIC14/handover in AUC. **(b)** Every pair: each point is one pair's share, the median over its 5 metrics (the rows of Table 4); the bar is the median over the 19 pairs; open circles are the two BPIC14 pairs, where analyst choices rather than resampling carry the variance. The medians in (b) do not add.

quantity — the median over pairs of the median over metrics of the largest within-metric first-order index — and cannot be read off the table's columns.

Most of the pooled surface's variance involves the evaluation split. Table 4 and Figure 3 partition the decomposition by (3), exactly and on every decomposition separately. Taking each column's median across the corpus — *four medians, which do not add, and are not four parts of one pair* — the components involving the split carry 71.7% of the variance, against 13.3% for the choices an analyst makes, 4.2% for the register-quality counterfactual and 4.5% for the components that join them. Resampling is the largest of the four on 17 of the 19 pairs. That share has a benchmark. With six split levels, the components involving the split hold 83.4% to 83.7% of the degrees of freedom, so a surface with no signal, whose cells differ only by independent noise, would put about that share there; the observed share is below it on 17 of the 19 pairs, and the analyst's choices carry 13.3% against the 2.3% such a surface would give them. The split stratum is

log	target	analyst choice	counterfactual	mixed	resampling
BPIC13_closed	handover	31.5	2.9	5.8	59.8
BPIC13_incidents	duration	13.3	2.9	3.0	79.6
BPIC13_incidents	handover	19.2	9.9	4.5	42.3
BPIC14	duration	71.9	15.1	7.9	5.1
BPIC14	handover	73.9	12.7	10.3	3.6
BPIC15_1	duration	9.4	3.9	4.7	80.5
BPIC15_1	handover	13.4	3.6	3.6	62.6
BPIC15_3	duration	11.7	8.1	1.5	77.6
BPIC15_3	handover	22.3	0.4	2.5	76.1
BPIC15_4	duration	11.3	2.5	5.6	81.6
BPIC15_4	handover	12.4	3.1	5.7	71.7
BPIC15_5	duration	17.4	4.2	3.9	69.9
BPIC17	duration	4.2	6.2	5.2	84.1
BPIC19	duration	3.8	1.1	2.8	90.9
Helpdesk	duration	31.7	8.6	1.7	53.6
RoadFines	duration	7.0	28.9	2.1	61.7
Sepsis	duration	13.2	14.5	5.0	71.9
UCI498	duration	11.0	3.0	4.3	81.1
UCI498	handover	41.0	7.1	9.3	42.4

Table 4: **The decomposition with the split as an error stratum.** Per cent of the pooled variance of V_s, per pair, median over the 5 metrics, under the equal-level measure. The four columns are the exact partition of Equation (3), which sums to 100 on every per-metric decomposition to within 4.4×10^{-16}; the printed medians need not. 'resampling' is every component involving the split, five of whose six levels are expanding-window folds.

where the cells' estimation noise lands, and a pooled surface that reads it as specification sensitivity overstates the forecaster's latitude; that is why the decomposition separates it. Resampling the pairs puts that median in [59.8%, 79.6%] — an interval over which pairs make up the corpus, not over sampling error within a pair — and it is similar across metrics: in AUC alone it is 68.4%, in Brier skill alone 69.9%. The split's main effect carries a median 18.9% and its interactions with the other axes 43.7%, so most of the stratum is interaction: effects of the other axes that change from fold to fold, which this design cannot separate from cell-specific estimation error. The fold-to-fold variation also trends in the fold index more than exchangeable folds would, which is

consistent with the time structure of an expanding window — a growing training set and a moving test period — and with serially correlated estimation error, which the design cannot tell apart (Supplement S14.7). Dropping the one split level that is a genuine analyst choice — the single temporal holdout, leaving five folds of one rule — the share falls only to 71.0%, so the stratum is not an artefact of mixing a rule with its folds. The two exceptions are the case study's own pairs, where the analyst axes carry the richest grids in the corpus — 4 pipelines and 10 register-quality conditions, against at most 3 and 6 on every other pair — and the ordering reverses.

Net of resampling, no single axis dominates — a weaker statement than the pooled surface suggests. On the fold-averaged surface (Table S12), which carries 28.3% of the pooled variance, the median first-order index is 21.9% for the baseline, 19.2% for the register-quality condition and 15.2% for the pipeline, and at the median pair the largest within-metric first-order index (a median over metrics) is 38.9%. The total higher-order share is 26.3% once resampling is removed, against 56.0% on the pooled surface. The higher-order share exceeds the largest within-metric first-order index on 13 of the 19 pairs before resampling is removed, and on 7 after. Counting the five folds as five specifications makes the interactions look dominant; treated as resamples, they are not. What remains is that which axis leads is a property of the pair — the baseline on 8, the register-quality condition on 5 and the pipeline on 6 — so an analyst cannot learn from somebody else's paper which of their own choices their answer turns on.

The pooled decomposition is reported beside it because it is what the specification-curve literature computes (Supplement S15.10). The measure is not neutral: across the three product measures the higher-order share moves by 24.3 percentage points, and under the measure concentrated on the reference cell the interactions no longer carry more than the largest first-order index.

The baseline moves the answer by more than the answer, on the pairs where an analyst has more than one baseline to choose between. Holding every other axis at the reference and varying only the baseline, the increment moves by 0.073 AUC at the median pair and 0.252 at the widest — larger than the increment itself on 15 of the 19 pairs. Without the half-of-intake baseline, which is impoverished in the way the excluded intercept-only baseline is, the spread is 0.0055 AUC at the median pair and 0.2495 at the widest, exceeding the reference increment on 7 pairs. On 17 of the 19 pairs only two baselines then remain, so that spread *is* the absorption D of the free field (Section 7.1; Supplement S15.17). The pipeline axis is itself two

axes, a model family and an encoding, and which of the two leads is again a property of the pair (Supplement S15.1).

6.3. Resolution regions, and the coverage they have

Table 3 carries the region labels and the robustness index for every pair (Table S21 and Figure S4 give the detail), computed on the inference surface under the whole-surface band at the empirical critical value. That family has 180 members on every pair and a median critical value of 5.03, against 2.37 for the within-cell family; under the narrower family the same draws give a different label on 12 of the 19 pairs, and resolve 1,100 cells against the whole-surface family's 698 (summed over pairs).

The multiplier approximation would resolve 890 cells at a median robustness index of 0.039 against 0.000 here, but it undercovers (Section 9.2): the 192 extra cells it would resolve are resolved below nominal coverage, so they are not counted here. Two shortfalls remain and are stated rather than corrected: the pointwise interval underneath the band undercovers where K/n is large — the median pair sits at 0.106 — and Supplement S9 reports the labels under an (n, K) widening as a sensitivity only; and the coverage measurement's bootstrap is ideal, so whatever the real resampling adds is outside it.

Of the 11 pairs that resolve anything, 9 resolve cells of one sign only — 7 beneficial and 2 harmful — and 2 resolve cells of both. That is the informative reading of the region column: on 9 of the 11 pairs where the data determine any sign at all, every sign they determine points the same way. Part of the both-signs reading is disagreement between metrics rather than across the surface: on BPIC19 the resolved cells share one sign within each metric — 8 beneficial in AUC, 25 harmful in Brier skill — and split by metric, the pooled band's cells would carry a different label on 8 of the 19 pairs in at least one metric (Table S7).

One surface of the 19 is uniformly beneficial, over a family that is 3.75% of that pair's admissible cells. $\rho = 1$ obtains only on that pair, the one carrying the richest declared grid, over an inference family of 180 cells against that pair's 4,800 admissible ones; a claim about every cell is easier to sustain over fewer of them, so the one case in which a single number is safe is a claim about 3.75% of the specifications an analyst could have chosen. The label is not an artefact of the critical value: it holds under the multiplier band, the reported band, that band's order-statistic upper end and every (n, K) widening (Supplement S9.7 and Supplement S15.16).

6.4. What one number costs

A conventional report disagrees in *sign* with a median 32.8% of the other admissible specifications under the equal-level measure. This rate overstates the risk of a one-number report, for three reasons that we apply cumulatively: not every axis is an analyst choice, most cells are not resolved, and many disagreements are too small to matter.

The axes are not all the same kind of thing. Table S25 sums first-order indices into *analyst latitude* (the pipeline and the baseline), *resampling* (the split) and the *counterfactual* (the register-quality condition), with the higher-order remainder a fourth column. Within each metric, analyst latitude carries a median 11.1% of the variance, resampling 18.9% and the counterfactual 4.2%, with resampling the larger on 10 of the 19 pairs; on a common headroom scale the ordering reverses, so neither ordering is a corpus fact independent of the scale (Supplement S3.7). Restricted to the analyst-latitude cells — register clean, split at the single holdout, a median 30 cells per pair — the median disagreement rate is 20.0%, and that is what a one-number report exposes a reader to. Across 26 choices of the cell called conventional, the all-cells rate runs from 32.8% to 38.4% (Table S26).

And most cells are not cells whose sign the data determine. The median pair resolves 2 cells of the 180 it carries, and sign agreement between two draws from a distribution centred near zero is one half by construction. **And a sign flip between two increments that both round to zero is not a misstatement anybody would act on**: 8 of the 19 reference increments are themselves smaller than 0.01 AUC in absolute value, so a *minimal practically important difference* is declared — MPID = 0.01 AUC — and a disagreement is counted only when the reference increment and the cell's own increment both exceed it, on the AUC sub-surface, because Remark 1 forbids carrying an AUC threshold to another metric (Table S32).

On the cells the corpus can resolve, the rate is 13.0%, and it is not an AUC statement. Of the 698 cells whose whole-surface band excludes zero, 91 disagree in sign with the reference cell. Split by metric that rate runs 2.6% on ROC AUC to 20.3% on log-loss skill: the pooled figure is carried by the metrics that are not the one the reference cell is read in, which is Remark 1 showing up in the corpus. Read against the reference cell of each cell's own metric instead, 20 of the 698 cells disagree, 2.9%; all 71 cells that leave the count are BPIC19's Brier, log-loss and Nagelkerke cells, on a pair whose resolved AUC cells are beneficial and whose resolved

Brier-skill cells are harmful (Supplement S14.1).

Imposing all three restrictions leaves 31 cells, of which 0 disagree. Restrict, on the AUC sub-surface, to the axes an analyst chooses, to the cells the band resolves, and to increments that clear the MPID at both ends. Only the 684 of the 5,808 admissible AUC cells that carry bootstrap draws can be restricted this way; 31 survive, and the rate is 0.0%. **We therefore do not claim that a one-number report misstates a sign at a rate worth quoting as a headline.** Two corollaries follow, and neither is comfortable for the conventional report. First, few cells remain because *the data resolve so little*: 155 of the 684 banded AUC cells resolve, 22.7% of them, and 14 of the 19 pairs do not resolve the reference cell itself. Second, the surface still moves, by a distance that dwarfs the MPID: over the cells the band resolves, the mean distance from the reference increment is a median 0.0600 AUC over the 9 pairs that resolve any AUC cell, and 0.0511 under the full restriction. The one-number report is seldom wrong about the sign; what it does not show is how far the rest of the declared surface lies from it.

6.5. Comparison with specification-curve analysis

Specification-curve analysis (Simonsohn et al., 2020) enumerates the reasonable specifications, plots the curve, and tests it against a sharp null by permutation. Run unchanged on every admitted pair (Supplement S18), it parts company with the region label on 9 of the 19 pairs: the permutation test rejects the sharp null and the same cells contain increments of *both* signs. The permutation test answers *is there anything here*; the region answers *may I quote a sign*, which is what a forecast user deciding whether to act needs, and the two should be reported together.

7. Case study: forecasting incident reassignment at a bank's service desk

This section evaluates one register against a named forecasting decision on a public log: will this incident be reassigned to another group before it is resolved? — a binary forecast issued at a declared origin. The log is BPI Challenge 2014 (van Dongen, 2014), from a bank's IT service desk; the register is its configuration management database, the inventory of the IT items incidents concern, and the register field is the affected configuration item, which takes 2,929 distinct values on the case-study cohort at incident creation. We report it in detail because it is the only log on which the forecast origin is varied, and there the origin decides the answer.

7.1. One log, two answers, and what separates them

This log appears twice. Section 6 applies the pre-specified generic rules of Section 5, which admit 46,606 cases and define the target by the pre-specified handover rule. This section uses the cohort and the reassignment target defined for the service desk: 45,455 cases at prevalence 0.400 (0.372 in the test half), 31,818 in training and 13,637 in test. The case-study cohort is a strict subset of the pre-specified one: the 1,151 cases the pre-specified rules admit on top of it are those the desk's own cutoff excludes.

The two give different answers about the register at the later origin, though not answers anybody would act on differently. Against intake plus group plus the knowledge reference, the case-study cohort and reassignment target give -0.0029 and the pre-specified cohort and handover target give $+0.00768$: opposite signs, both inside the 0.01 AUC that Section 6.4 declares the minimal practically important difference. On the second, the pre-specified contrast that the register adds at the later origin (PC3, Supplement S3.5) has a one-sided p-value that rejects at the Holm-adjusted level while its interval does not exclude zero; the interval decides, and the supplement says why the two disagree. Table S15 reconciles, one factor at a time, the value an earlier, unpublished analysis of this log reported with the pre-specified one.

It is the target, not the cohort. Over the cohort $\times$ target design the target carries 99.3% of the variance in the increment, the cohort 0.0%, and their interaction 0.7%. Reassignment to another group before resolution has prevalence 0.400 over the case-study cohort; handover between activity groups, under the pre-specified rule, has prevalence 0.927 over the pre-specified cohort. They are not the same question and the register is not worth the same to them. It is not a data-preparation difference — the two analyses reach the same attribute values on every row — nor a modelling difference, since they run the same pipeline at the same reference cell.

The tie-break is declared. 270 incidents share their timestamp with another, so the temporal split at seventy per cent of the row order depends on a tie-breaking rule; over 25 random tie-breaks the increment ranges over 0.00088 AUC without changing sign, and every number in this section uses a declared tie-break by incident identifier (Supplement S15.13).

The absorption comes first. The absorption D of (2) is reported with its own interval and the ratio R second, with a Fieller set: at the reference cell D is $+0.0804$ [$+0.0626$, $+0.0914$] and R is 0.437 with a bounded Fieller set (Supplement S20).

forecast origin	baseline	n	prevalence	register levels	base AUC	with AUC	V	interval	resolved
τ_0 first touch, every call	intake	145,478	0.146	4,034	0.554	0.785	+0.2310	[+0.2204, +0.2618]	yes
τ_1 first touch, escalating only	intake	44,513	0.402	2,445	0.552	0.754	+0.2013	[+0.1930, +0.2224]	yes
τ_2 incident creation	intake	45,455	0.400	2,929	0.562	0.746	+0.1838	[+0.1725, +0.2040]	yes
τ_2 incident creation	intake + group	45,455	0.400	2,929	0.644	0.748	+0.1034	[+0.0949, +0.1295]	yes
τ_2 incident creation	intake + group + knowledge	45,455	0.400	2,929	0.805	0.802	−0.0029	[−0.0074, +0.0019]	no
τ_2 incident creation, τ_1's population	intake	44,513	0.402	2,887	0.559	0.750	+0.1914	[+0.1832, +0.2135]	yes

Table 5: The forecast origin implemented as an axis. Each origin carries its own population, register and admissible set; the forecast target is the same at all of them. τ_1 and τ_2 differ in what is known and in which cases are in scope, because 942 incidents have no interaction record; the sixth row is τ_2 restricted to τ_1's own population, and it is that row, not the third, that isolates the information step. Every row is on the case-study cohort and the reassignment target, under a declared tie-break; intervals are pointwise basic at 300 draws.

7.2. Three forecast origins

A forecast origin fixes three things at once — which cases are in scope, what is known about them, and which fields are admissible — and the case-study surface is estimated at three origins.

The register's worth turns on which origin the forecast is issued at: at τ_0 it is +0.2310 [+0.2204, +0.2618] over 145,478 *interactions* — calls to the desk, of which only a minority ever become incidents, which is why that population is far larger than the 45,455 incidents the later origins carry — at τ_1 +0.2013 [+0.1930, +0.2224], and at τ_2 against the intake block plus the group it is +0.1034 [+0.0949, +0.1295] while against the block, the group and the knowledge reference it is −0.0029 [−0.0074, +0.0019] — not resolvably different from zero. The step from τ_1 to τ_2 is an information step only on the matched population: 942 of the 45,455 incidents have no interaction record, so the sixth row of Table 5 restricts τ_2 to τ_1's own cases. Read as steps, the increment moves by −0.0297 AUC from τ_0 to τ_1, which is the population; by −0.0099 from τ_1 to τ_2 on the matched population, which is the information; and by −0.0076 more from the matched population to the full one, which is the incidents without an interaction record (Supplement S15.13).

Two further experiments are in Supplement S15.5. The first is an information-set (look-ahead) sensitivity: writing λ for the share of the knowledge reference's values assumed written *after* the incident, the increment reaches half its knowledge-free value at $\lambda = 0.64$, so an unprovable admissibility assumption is reported as a sensitivity analysis and not as a headline. The second is a register-quality axis under the six declared mechanisms and, beside them, three dependent mechanisms and a discovery lag, applied to both halves of the data with their parameters estimated from training alone (Supplement S11).

8. Decision-analytic evaluation

An increment in AUC is not a quantity a service desk can act on. This section reads the same surface in net benefit — the decision-based value of a forecast in the sense of Murphy (1993) and Granger & Pesaran (2000) — which is the number of correctly nominated cases per thousand arrivals, net of the false nominations, at a declared exchange rate between the two.

8.1. The operating point is a cost ratio, and calibration comes first

At threshold θ the net benefit of a rule that nominates when $\hat{p} \geq \theta$ is

$$\mathrm{NB}(\theta) \;=\; \frac{\mathrm{TP}(\theta)}{n} \;-\; \frac{\mathrm{FP}(\theta)}{n} \cdot \frac{\theta}{1-\theta},$$

so $(1-\theta)/\theta$ is the number of false nominations the desk accepts for one true one, and choosing θ is choosing an operational cost ratio that belongs to the analyst's declaration rather than to a grid search. We report the whole grid — 31 points from 0.05 to 0.80 — and name three points a desk might occupy: $\theta = 0.10$ (9.0 false nominations per true one, a cheap review), 0.30 (2.3, a review and a miss costing about the same) and 0.60 (0.7, scarce review capacity); a reader whose desk sits elsewhere reads their own point off Figure S3. A threshold is a statement about probability, so every decision curve here is computed on *calibrated* scores, with the calibrator fitted inside the training half and rebuilt inside every bootstrap refit, and a pre-specified rule excludes an arm whose calibration slope leaves $[0.5, 2]$ rather than repairing it, because recalibration is already a level of the pipeline axis (Supplement S15.6).

8.2. Simultaneous bands change what can be said

The decision-curve family is declared separately from the scalar surface and carries its own max-t critical value, because net benefit and a scalar metric are not the same scale. Its critical value is the empirical quantile of Section 4.3, taken over the *admissible* family: a cell whose net-benefit difference is exactly zero in at least nine draws of ten is zero by arithmetic, not by estimation, and is excluded before the quantile is taken because it is not a statement about the register while its vanishing spread inflates the value every other cell is judged by (Section 9.2). On the case-study family that rule excludes 2 of 248 cells and gives a critical value of 6.64 [5.73, 9.08], against 3.57 from the multiplier approximation. The empirical band is the better of the two but remains about

a percentage point short of nominal on its matched family, which fits this corpus less closely than the surface families do (Section 9.2).

Across the 31 operating points of the case-study curve **the pointwise interval declares the increment resolvably positive at 24 of them and the simultaneous band at 8**, the lowest threshold the second resolves being $\theta = 0.60$; the harmful counts are 0 against 0. All are built from the same draws, so what differs is the family and the level it is bought at: those points are the price of a statement about a *range* of thresholds rather than about each one separately. Supplement S15.7 gives the counts under the multiplier, under the order-statistic upper end of the critical value and with the excluded cells kept, none of which changes the conclusion; Figure S3 draws the curve (Table S30 is a score-threshold sensitivity analysis on raw scores, not this curve). The pointwise interval is not widened by a family-wise factor, and no (n, K) widening is applied to this family.

8.3. The decision, in units a service desk uses

The comparison is run in net benefit per thousand cases, over a declared distribution of exchange rates a desk plausibly holds (0.08 to 0.36, centred 0.19), and only on models that pass the pre-specified calibration rule — which excludes 40 of 118 models, a model being a pair–pipeline–baseline triple whose two arms must both pass, and 3 of 19 pairs. This is the most consequential operational finding: a decision curve read off a model whose probabilities are wrong is read at a threshold the model does not have.

The four decision rules separate on most pairs, with the excess falling on the conservative rule rather than the one-number rule (Supplement S15.9). Larger than anything the choice of decision rule costs a one-number report is which fields the forecast origin makes admissible: on the case-study log's pre-specified handover target, the register promises 1.71 per thousand against the arrival attributes alone and delivers 0.003 once the assignment group and the knowledge reference — assumed recorded by incident creation — are in the benchmark, a shortfall of 1.70. Supplement S15.9 carries the rules, the exchange-rate distribution and the exclusions; Tables S28 and S34 the numbers.

9. Simulation against a known answer

Everything so far has been measured on real logs, where the answer is not known. This section measures the interval and the band where it is, against worlds whose population increment is computed in closed form. The full tables are in Supplement S16 and the diagnosis of the one world that failed is in Supplement S6.

9.1. What the simulation certifies of the pointwise interval

The simulation uses six worlds — four of them misspecified for the estimator — at 1,000 replicates each, with the covariate space finite so that the population AUC of any score function is computed in closed form. Two estimands are separated because under misspecification they differ: V_{oracle}, the increment between the two Bayes-optimal rules, and V_{limit}, the increment between the two fitted pipelines' limits (the simulation resamples by subset moving blocks, whereas the intervals and bands the article reports draw Dirichlet weights per block, Section 4.1, so what is certified here is the subset construction, and Supplement S3.9 compares the two schemes on the corpus's own design); a resampling interval can only cover the second (Supplement S6.1, Table S44). Coverage estimates carry a Monte Carlo standard error of about 0.7 percentage points, which is the precision every comparison here should be read at; Figure S5 draws them by construction and world.

Three things are established and one is not. The nested basic interval covers a median 93.0% against the fixed-model percentile interval's 92.1%, and on the sparse world the *percentile* interval collapses to 37.2% while the basic construction restores it to 89.9%, which is why every interval and band here is the basic one (Supplement S6.2). Against V_{oracle} every construction undercovers on the drifting world — 65.8% at best — and it should, because when the coefficients move between eras the estimator is not consistent for the oracle quantity, and no resampling scheme repairs that. Varying sample size and register cardinality jointly on a grid of 10 (n, K) cells, up to 8,000 training rows and $K = 3{,}019$ levels (Table S45), the ratio K/n governs the basic interval's coverage — a median 83.3% and a minimum 21.3% across that grid, short of nominal everywhere — and does not exhaust it: holding the register at one cardinality while the training half grows from 2,000 to 8,000 rows moves coverage from 0.453 to 0.613 (Supplement S9). This corpus sits in the regime where that matters, its median pair at $K/n = 0.106$ with 10 of 19 pairs above a tenth, so a nominal pointwise interval on them is narrower than it should be; past a crossing in n the basic interval stops containing its own point estimate (Supplement S9.4). The block length, varied over 3 settings

around $\lceil n^{1/3} \rceil$, moves coverage by up to 3.6 percentage points within a world without changing any comparison the paper rests on.

9.2. The band's family-wise coverage

Everything above validates a *pointwise* interval. Two of this paper's three reporting objects are functions of a *simultaneous* band, and this section measures what the band's critical value covers.

The design, and why its answer is an upper bound. One draw from the sampling distribution of a synthetic family gives the point estimate, and as many further draws from that same distribution as the matched corpus family carries give the bootstrap, so the bootstrap is the sampling distribution itself: every way a resampling scheme can be wrong is switched off, and a coverage measured here is the most the real procedure can attain. The families are the ones the corpus runs — every pair's surface family of 180 cells at 400 draws, the corpus's modal decision-curve family and the case study's own curve family (Table S47) — matched in cross-cell correlation, effective rank, per-cell excess kurtosis and the ratio of the empirical quantile to the multiplier on the corpus's own families. The surface family reproduces that ratio at 1.25 against the corpus's median 1.50, inside its interquartile range; the curve family reaches 1.49 against 2.24, outside it, so its numbers below transfer less well. Four regimes are run: Gaussian draws, the case the multiplier is derived for; draws carrying this corpus's tails; the same with the degenerate cells described below; and a fixed heterogeneous *non-zero* truth, under which a cell can be resolved correctly and family-wise coverage is the quantity a region label actually depends on. Every coverage here has a Monte Carlo standard error of at most 1.0% (Supplement S9.6).

The empirical quantile attains its level on the matched family; the multiplier does not, and no draw budget makes it. On every pair's surface family, under this corpus's tails, the empirical quantile of the observed maxima covers 94.7% against a nominal 95%, at a Monte Carlo standard error of 0.5%, and 94.6% under the non-zero truth; the order-statistic upper end of the same quantile covers 96.7%. The Gaussian multiplier covers 91.5% and 92.6% on the same families — four to six standard errors short — and the two change places under the Gaussian draws it is derived for (Supplement S3.6). Across draw counts at the surface family the multiplier rises steeply out of the smallest budgets and then *plateaus*: 80.6% at 33 draws, 90.9% at 150, 91.5% at 400 and 91.2% at 1,000. The empirical quantile behaves differently over the same counts — 77.8%, 93.8%, 94.7% and 94.4% — because an order statistic of 33 maxima is badly determined and one

of 400 is not, which is why the construction the paper reports is available at this corpus's draw count and would not have been at a third of it. What the level costs is width: the empirical band is 1.24 times as wide as the multiplier's (the mean band-width ratio of this run), and Section 6.3 prints what each resolves.

The decision-curve families, and the cells that are zero by arithmetic. Where both arms treat every case alike the net-benefit difference is zero by arithmetic; 638 cells of the corpus's curve families are exactly zero in at least nine draws of ten, and standardising such a cell by its vanishing spread inflates the critical value every other cell is judged by, so a declared rule removes such cells from a decision-curve family before the quantile is taken (Supplement S9.6). On the modal curve family so treated the empirical quantile covers 93.8% under the corpus's tails against the multiplier's 91.3%, and on the case study's own family 93.6% against 91.2%. **On the curve families the empirical quantile is therefore about a point short of its level**, on a synthetic family that under-reproduces the corpus's estimator ratio, and it is reported as the better of the two estimators rather than as one measured at nominal.

What the simulation does not establish. It certifies the pointwise constructions on six worlds, a grid of 10 (n, K) cells and 3 block lengths, and measures the band's critical value on the 3 matched families of Section 9.2 under four regimes — on nothing else. The mixing the moving-block bootstrap needs is assumed and not tested, and the stationarity it also needs is tested (Section 3.2) and does not hold on every pair; the split point is fixed within a draw, so split-position uncertainty is outside every interval reported here; and the drifting world shows what happens when the estimator is not consistent for the quantity a reader cares about. The (n, K) widening that addresses the pointwise channel is fitted pointwise, applied simultaneously and measurably misspecified in n, and is reported as a sensitivity only (Supplement S9).

10. Threats to validity

Construct — this is forecast performance, not value. Nothing here measures what a register costs to acquire or maintain, whether acting on a forecast changes an outcome, or what a data programme is worth. The corpus is a family of related questions rather than replications — the two pre-specified targets label the same case identically on a median 51% of cases, and the registers run from a configuration item to a legal article — so every corpus statistic is about *sensitivity*, every one is

reported on the 8 ITSM pairs as well (Table S22), and Table S17 names each pair's own construct threat. The pre-specified handover target on the case study's log is close to degenerate at prevalence 0.927 and is where Section 7.1 shows the register's sign turns; whether a generic handover rule is a meaningful target on ITSM logs and whether the prevalence window should close below 0.95 are open, and the rule was not narrowed after seeing which pair it admits. The quality mechanisms are applied to both halves with parameters estimated from training alone (Section 7.2), so they are the operational counterfactual and not a train–test mismatch, but they remain constructed, and validating them against a genuinely incomplete register is the obvious next study.

Internal — what the band's coverage does and does not establish. Section 9.2 measures family-wise coverage over matched families whose bootstrap is *ideal*, so every coverage there is an upper bound on what the procedure attains, and the empirical critical value the region labels rest on attains its level there while the multiplier approximation does not. Three qualifications are limits rather than results — the K/n-governed undercoverage of the pointwise interval underneath the band, the width at which the level is bought, and a decision-curve critical value about a point short of its level — and Supplement S21 states them, with the mechanisms this evidence rules out and the further assumptions it names.

Internal — the estimand, and nested comparisons near zero. Definition 2 defines the increment as a large-sample limit of the fitted rule, while the split axis, the refitting bootstrap and the forecast origins treat the forecasting *method*, with its finite estimation window, as the object, as Giacomini & White (2006) do. Near a zero increment the two differ: estimating the larger model's extra parameters biases an out-of-sample comparison against it (Clark & West, 2007), so some harmful labels on proper scores may be finite-sample effects rather than evidence against the predictor. At the reference cell a Clark–West comparison favours the predictor on 12 of the 19 pairs against 7 for the raw Brier comparison, counting one-sided rejections in the predictor's favour at the 5% level, and 3 of those pairs have a negative raw Brier difference (Table S9). A surface-wide version of that comparison, a zero-increment world in the simulation of Section 9, and bootstrap weights shared across splits — which would let sampling noise be subtracted from the split stratum — each need a new bootstrap run and are left open.

External — the design space is declared, not complete. $\mathcal{S}$ has nine axes in Definition 1 and the per-pair factorial varies 5 of them (Table S20); the forecast origin varies on the case study's log alone, the operating point only inside the decision-curve family, the cohort only in the case study's cohort design, and the target indexes surfaces rather than varying within one. It could have more — tuning protocol, missing-data treatment, target horizon, label-noise model — so every claim here is relative to this space. Prefix length, prefix bucketing and sequence encoding are not axes, because this study forecasts at one origin from creation-time attributes, so no result here speaks to benchmarks that vary them. Forecast origins later in a case's life are a finer grid of the forecast-origin axis, and a pilot on one log (Supplement S12) shows that they matter: holding the register fixed and moving only the prefix, the increment falls from +0.257 AUC at case creation to +0.064 two events later and +0.069 at the longest prefix run, and is undetermined in sign at 2 of 6 prefixes. Case creation — the origin every other result here occupies — is thus the most favourable prefix for a creation-time register, and two studies reporting from the two ends of that axis would differ by a factor of 3.7 without either being wrong. One log, one register and no sequence encoder is a pilot, and the omission stands.

External — narrower limits. The inference surface is smaller than the declared one and always will be, because a draw refits every arm of every cell, so every region label is a claim about a fraction of the space the point estimates cover, on which *uniformly beneficial* is easier to earn (Section 4.2), and every label stays conditional on the register-quality axis being averaged over. The pipeline axis is crossed on the 8 ITSM pairs and not elsewhere, and what the crossing found is not a clean replication (Supplement S15.1), so the case study's ordering is the common case and not a corpus fact. The case study is one bank's service desk on one public log, with the availability question at its centre settled by probability rather than by a timestamped field-value history. And reporting practice is not measured (Supplement S24).

11. Conclusion

What a predictor adds to a forecast is a functional of design choices, not a number. This paper made that operational: a declared design space, a surface over it, an exact decomposition of the surface's variance that separates the choices an analyst makes from the resampling the design also contains, simultaneous bands whose family-wise coverage is measured in a matched

simulation rather than assumed — an upper bound, since that simulation's bootstrap is ideal, and the pointwise intervals beneath the bands undercover where the predictor has many levels relative to the sample — and region labels that say where a sign is determined and withhold a direction where too little of the surface resolves to support one.

On 7 of 19 pairs the baseline alone — among the baselines an analyst would actually build — moves the increment further than the increment itself.

Most of a pooled specification surface is resampling, and net of it different pairs turn on different analyst choices. The components involving the split carry a median 71.7% of the variance, below the 83.7% a surface with no signal would put there. Net of them the largest first-order index at the median pair is 38.9% against a higher-order share of 26.3%, and the interactions exceed the largest within-metric first-order index on 7 of 19 pairs, against 13 before the resampling stratum is removed, so once the folds are treated as resamples the interactions do not dominate (Section 6.2). Which axis leads is a property of the pair — the baseline on 8 pairs, the register-quality condition on 5 and the pipeline on 6 — so an analyst cannot learn from somebody else's paper which of their own choices their answer turns on.

A one-number report is usually right about the sign and usually silent about the size. Once a sign has to be determined by the data *and* worth acting on at 0.01 AUC, the number of the 31 AUC cells surviving all three restrictions that contradict the conventional report is 0, **so the sign-disagreement rate is not a headline** (Section 6.4). What the conventional report does not say is that the cells the band *resolves*, on the 9 pairs that resolve any, lie a median 0.0600 AUC away from the one it stands on, and that on 14 of 19 pairs the data do not determine the sign of its own cell at all.

The forecast origin is an argument of the estimand, not a preliminary. On the case study, moving the forecast from first touch to incident creation moves the increment by -0.0297 AUC through the population, by a further -0.0099 through the information once the population is held fixed, and by -0.0076 more through the incidents that have no interaction record — before any baseline changes; and the same log's answer changes sign when the target changes, within the minimal practically important difference, with 99.3% of the difference on the target and 0.0% on the cohort. A desk that adopted this register on its estimate against the arrival attributes alone, and deployed it where the assignment group and the knowledge reference are already recorded,

would be promised 1.71 net true positives per thousand cases on the pre-specified handover target and receive 0.003.

And a threshold is not always a cost ratio. On 40 of 118 models — pair–pipeline–baseline triples — the calibration slope falls outside $[0.5, 2]$, so the threshold at which a decision curve is read cannot be interpreted as the cost ratio the decision-analytic argument requires (Section 8.1).

What to report. The design space and the measure over it; which of the three sets of specifications a claim quantifies over; the absolute increment at every baseline rather than a ratio; the decomposition with the resampling stratum separated and $1 - \sum_i S_i$ reported as such; the region and ρ under the band's own family, with the counts they rest on; the forecast origin and the target as parts of the estimand; a minimal practically important difference declared before any disagreement rate; and the reduction R last if at all, with a Fieller set and only where its denominator is resolvably positive. Supplement S22 states the standard in full, with a six-step recipe, the accompanying software package (Anonymous, 2026) that implements the objects, and a worked example on a dataset that is not an event log.

Data availability

All data are public, and every admitted event log is cited by its DOI: BPI Challenge 2014 (https://doi.org/10.4121/uuid:c3e5d162-0cfd-4bb0-bd82-af5268819c35; van Dongen, 2014), BPI Challenge 2013 incidents (https://doi.org/10.4121/uuid:500573e6-accc-4b0c-9576-aa5468b10cee; Steeman, 2013b), and Steeman (2013a), van Dongen (2015), van Dongen (2017), van Dongen (2019), Polato (2017), de Leoni & Mannhardt (2015), Mannhardt (2016) and Amaral, Fantinato & Peres (2018); the SHA-256 checksum of every downloaded file is listed in the checksum list in the archive. 26 files are downloaded and checksummed in all, one log shipping as three; the pre-specified role-assignment rules of Section 5 admit 13 logs, yielding the 19 log–target pairs analysed here. Table S16 accounts for every other downloaded log, either by the exclusion code the rules gave it or by the code HELD_OUT, which marks a log reserved for a separate pre-specified prediction and never offered to the admission rules (1 of the checksummed files). Every analysis result is in the release named in the code-availability statement, and one command regenerates every result from the public logs; where later code would print a summary differently, the correction register (Supplement S23) says so.

Code availability

Code, derived results and the pre-specified protocols with their amendment log are archived under a persistent identifier and developed in a public repository; both are given on the title page and are omitted here for double-blind reviewing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Anthropic Claude (`claude-opus-5`, `claude-fable-5`, `claude-fable-5-1` and `claude-opus-5-5`; 2026-08-18 to 2026-09-28; identifiers recorded from 2026-08-24, with gaps the log states) in order to draft and edit prose, to write analysis and verification code, and to conduct internal critical reads of the manuscript against its code and results. After using this tool the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

The use in writing code is described in the Methods section (Section 4.1). No generative model produced, imputed, augmented or selected any datum, any result or any citation behind a claim of this paper: every result is produced by archived code that one command re-runs, and every reference was checked against its published record. Every session's date, and its model identifier where one was recorded, are in `AI-USE.md` in the code repository. The one place where a machine-assisted judgement enters a study is the literature pilot of Supplement S24, which is declared where it is described, and no claim in this paper rests on it.

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